



GITAM School of Business

Visakhapatnam

MBA (2024-26)

Title of the Project:

Smart City Water Project – Pune by Elite Hydro Constructions

as a part of the fulfilment of the Course

24OPSC7001 – Project Management

Term-V, October, 2025

Submitted By: Team 3

1. **Niteshwar Sai 2024037862**
2. **Vellore Rohit Ajrun Raj 2024035316**
3. **Bhavya Karri 2024175064**
4. **Sollem Ramesh 2024002458**
5. **Sirisha Varahal 2024075052**

Submitted to:

Dr Prasad Revur

Table of Contents

1.1 Background of the Smart City Mission.....	1
1.2 Description of Pune Smart Water Supply Project.....	1
1.3 Project Aims and Scope.....	2
1.4 Stakeholder Identification and Analysis.....	2
1.5 Project Constraints and Assumptions.....	3
2. WORK BREAKDOWN STRUCTURE (WBS)	4
3. Project communication, Procurement & Implementation Plan:	16
3.1 Communication management plan:	16
3.2 Procurement Strategy and Vendor Management:	17
3.3 Role of IoT and SCADA Systems:	17
3.4 Coordination Challenges Between IT and Civil Services:	18
3.5 Data Integration and Cybersecurity Risk mitigation:	18
4.1 Project Monitoring and Controlling Plan	19
4.2 Quality Control and Testing Procedure.....	19
4.3 Delays Due to Public Protests and Road Closures – Mitigation.....	20
4.4 Lessons Learned and Recommendations.....	20
4.5 Conclusion.....	21
References:	21

1.1 Background of the Smart City Mission

The Smart City Mission, initiated by the Government of India in 2015 under the Ministry of Housing and Urban Affairs (MoHUA), is one of the most ambitious urban renewal initiatives in the history of the country. The core aim of the mission is to enable inclusive and sustainable cities that offer basic infrastructure, a good quality of life to residents, and a clean and sustainable environment through the use of smart solutions and digital technologies. 100 Indian cities were competitively chosen through a process to be provided with financial support and technical assistance for smart projects in areas ranging from water management, waste management, to energy efficiency and transportation (MoHUA, 2021).

The mission lays stress on citizen engagement, collaborative planning, and technology-led innovation. Every city develops a Smart City Proposal (SCP) for area-based and pan-city plans. The focus domains are digital infrastructure, smart urban mobility, smart government, and particularly sustainable water management, which is seen as an essential pillar of urban resilience.

India's urban water networks have traditionally experienced problems such as leakage, unauthorized connections, and non-revenue water (NRW) losses running frequently over 35% in major metropolitan areas. The Smart City Mission accordingly takes precedence the upgrading of water facilities with Internet of Things (IoT) sensors, Supervisory Control and Data Acquisition (SCADA) systems, and Geographic Information Systems (GIS) for observation and data-based decision-making.

Pune, as a fast-growing city and education and industry hub, was chosen under the first phase of the mission. Its addition was intended to develop a model of smart urban water management that other cities could replicate. The Pune Smart City Water Supply Project therefore became a model scheme in the national effort, dovetailing with the Sustainable Development Goals (SDG 6: Clean Water and Sanitation) as well as India's Vision 2030 for inclusive urbanization.

1.2 Description of Pune Smart Water Supply Project

The Smart City Water Supply Project – Pune is among the most extensive and technologically intensive projects under Pune's Smart City Mission. Headed by the Pune Municipal Corporation (PMC), the project offers 24x7 round-the-clock water supply to around 150,000 residences in Pune's Zone III. The project duration is 36 months at an estimated cost of ₹300 crores, financed jointly by the Central Government, Maharashtra's Government, and PMC.

The project was undertaken due to the difficulties of intermittent supply, excessive leakage, and unequal distribution of water. Prior to the project, there was usually only 4–6 hours of water supply available in most places, with poor pressure control and manual metering systems. This inefficiency resulted in excessive water losses and public discontent.

To fill these gaps, the project incorporates IoT-based smart metering infrastructure, SCADA control, and auto-data collection platforms for real-time monitoring of water flow, pressure, and consumption. Pipeline replacement, zonal distribution networks, and leakage detection systems are also part of the project. In addition to this, PMC partnered with Elite Hydro Constructions and various IoT service providers to deploy end-to-end automation.

The system facilitates real-time data visualization from a centralized command center, allowing it to support predictive maintenance and rapid fault detection. This reduces unaccounted water losses

and enhances billing accuracy. Once rolled out in its entirety, the project should see a 30–35% reduction in NRW and an overall efficiency improvement of 40%, setting a standard for other Indian cities.

1.3 Project Aims and Scope

Aims

- The Pune Smart City Water Supply Project was envisaged with the following aims:
- To provide equitable 24x7 water supply to all categories of consumers in the specified smart zones.
- To minimize water losses through leak detection and optimization of networks through IoT and SCADA technologies.
- To enhance billing transparency by the installation of smart meters with real-time recording and automated billing.
- To create a central monitoring system for real-time observation and data analysis.
- To encourage people's participation and awareness through online platforms and mobile apps.
- To enhance institutional capacity by equipping municipal workers and contractors with smart water management technologies.

Scope

- The scope of the project goes beyond the construction of infrastructure—it includes the entire lifecycle of smart water governance. The principal components are:
- Infrastructure Modernization: Upgradation of old pipelines and installation of smart meters.
- Technology Integration: Installation of SCADA, IoT sensors, and GIS for real-time visibility of the system.
- Capacity Building: Training sessions for PMC staff, contractors, and local engineers.
- Data Analytics: Dashboards and mobile application development for monitoring performance.
- Public Communication: Involving citizens to report leaks and monitor supply through digital channels.

The project site is around 70 square kilometres in Zone III, with both residential and commercial areas. Attention is drawn to developing a replicable model that is scalable to other areas of Pune and ultimately to other smart cities.

1.4 Stakeholder Identification and Analysis

Good stakeholder management is essential to the success of any large urban infrastructure project. The Smart City Water Supply Project has several stakeholders with varying interests and influence.

Stakeholder	Role/Responsibility	Interest Level	Influence Level
Pune Municipal Corporation (PMC)	Project owner; funding, coordination, and monitoring	High	High
Smart City Mission Office	Policy guidance and financial oversight	High	High
Elite Hydro Constructions	Main civil contractor for pipeline works and smart infrastructure	Medium	High
IoT Service Providers	Supply and integration of smart meters, sensors, and control software	High	Medium
Community Representatives	Represent citizen concerns and feedback	High	Medium
State Urban Development Department	Regulatory approvals and funding channelization	Medium	High
Consultants and Auditors	Technical and financial evaluation	Medium	Medium
Local Citizens	End users of water services	High	Low

This matrix indicates that the PMC and Smart City Mission Office have high interest and high influence, so they are critical stakeholders. Contractors and IoT service providers are critical to operational implementation, whereas citizens' acceptance guarantees long-term viability for the project.

Participation of stakeholders involved town hall meetings, grievance redressal systems online, and special communication officers to manage public feedback and resolve issues during implementation. All this inclusive participation cut down opposition during excavation and metering installation.

1.5 Project Constraints and Assumptions

Constraints:

- All projects function under a set of limitations that influence its scope, cost, and duration. The key constraints identified in the current project are:
 - **Budget Constraints:** With a limit of ₹300 crores, there was a need to prevent cost overruns by following stringent procurement and scheduling.
 - **Time Pressure:** The 36-month project timeline necessitated following rigorous timelines irrespective of monsoon breaks.
 - **Coordination Challenges:** Merging civil and IT elements necessitated ongoing coordination among engineering and technology vendors.
 - **Geographical Spread:** Spanning 150,000 homes over 70 sq. km created logistic issues in installation and testing.
 - **Public Resistance:** Public inconvenience and protest in certain locations resulted from temporary road closures and trenching in connection with pipeline replacement.

Assumptions

- The planning process involved a number of assumptions to simulate risk and schedule viability:
- Timing of sufficient funding would be received by all the funding partners.
- SCADA and the IoT systems would complement existing municipal databases.
- Weather conditions at the local level (especially monsoon seasons) would result in manageable delays.
- Locally, there would be skilled technical manpower to support the maintenance of the system after implementation.
- Public support and awareness programs would reduce protests and resistance.
- These assumptions informed the initial project charter but were regularly reassessed during the implementation phase through risk assessment and stakeholder feedback processes.

2. WORK BREAKDOWN STRUCTURE (WBS)

WBS PHASE	NAME	START DATE	FINISH DATE	DURATION (DAYS)	KEY DELIVERABLES/ FOCUS
1.0	Project Initiation & Planning	03-11-2025	17-12-2025	33	Finalizing scope, approvals, and site survey.
2.0	Design & Engineering	18-12-2025	26-02-2026	51	Hydraulic, SCADA, IoT architecture, and civil drawings.
3.0	Procurement	27-02-2026	11-05-2026	52	BoQ preparation, vendor selection, and procurement of all civil/ IT hardware (pipes, meters, SCADA, servers).
4.0	Construction & Installation	12-05-2026	07-12-2026	150	Physical work: excavation, pipe laying, construction of chambers/ reservoirs, and

WBS PHASE	NAME	START DATE	FINISH DATE	DURATION (DAYS)	KEY DELIVERABLES/ FOCUS
					household meter installation.
5.0	System Integration (IoT & SCADA)	08-12-2026	12-04-2027	90	Configuring sensors, SCADA software setup, network connectivity, and integration with the billing system.
6.0	Testing & Commissioning	13-04-2027	28-07-2027	77	Pressure/ leakage testing, meter calibration, system performance testing, and pilot runs.
7.0	Training & Handover	29-07-2027	05-04-2028	180	Staff training, documentation, asset handover, and citizen awareness programs.
8.0	Monitoring & Evaluation	21-10-2027	25-04-2028	134	Continuous monitoring, maintenance scheduling, and final performance audit.

Now, let's look at the Phase breakdown for each individual phase

Phase 1.0: Project Initiation & Planning

- WBS ID: 1.0
- Duration: 33 Days

- Start Date: November 3, 2025
- Finish Date: December 17, 2025
- Objective: To officially launch the project, secure buy-in, and establish the foundational data and plans necessary for all subsequent phases.

WBS ID	Task Name	Duration (Days)	Key Activities & Deliverables
1.1	Define project objectives and scope	2	Formalizing the project charter and confirming core goals (24x7 supply, smart metering, leakage detection).
1.2	Identify stakeholders and form project team	3	Appointing key roles (PM, APMs, lead engineers) and establishing the project organization structure.
1.3	Approvals from PMC and Smart City Mission	3	Obtaining official sign-off on the charter, scope, and initial budget from the governing authorities.
1.4	Site survey and baseline data collection	14	Collecting GIS data, hydrological measurements (flow, pressure), and current non-revenue water (NRW) figures for Zone-III.
1.5	Prepare project schedule and budget plan	14	Developing the master Gantt chart, CPM analysis, and finalizing the detailed ₹500 Crore budget allocation per WBS element.

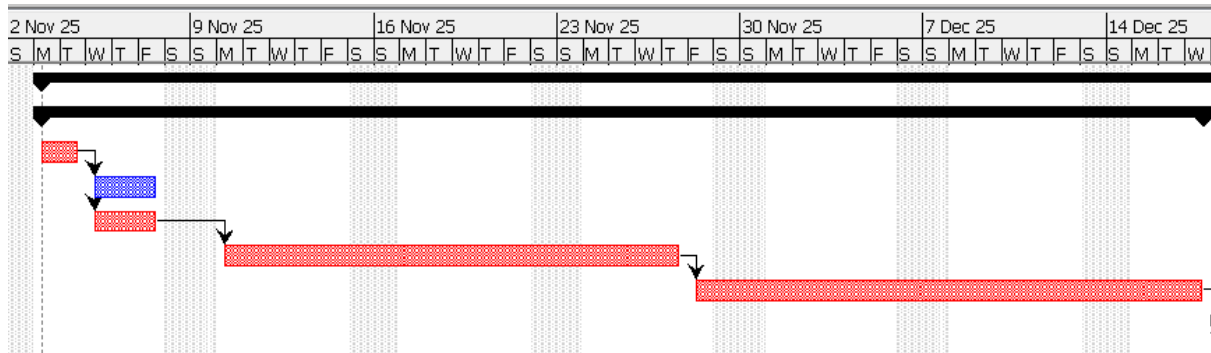


Figure 1: Phase-1.0 Breakdown

Phase 2.0: Design & Engineering

- WBS ID: 2.0
- Duration: 51 Days
- Start Date: December 18, 2025
- Finish Date: February 26, 2026
- Objective: To translate the project requirements into comprehensive technical specifications, blueprints, and digital architecture.

WBS ID	Task Name	Duration (Days)	Key Activities & Deliverables
2.1	Hydraulic and network design for Zone-III	30	Modelling the water distribution network, pressure management zones (DMAs), and determining required pipe diameters and valve locations.
2.2	Smart meter and IoT system architecture	30	Designing the communication network (e.g., LoRaWAN, Cellular) for 150,000 meters and selecting sensor types and placement strategies.
2.3	SCADA system design (data control & automation)	30	Designing the centralized control system, including data acquisition servers, control logic, and human-machine interface (HMI) screen layouts.

WBS ID	Task Name	Duration (Days)	Key Activities & Deliverables
2.4	Civil infrastructure drawings (pipeline layout, valves, tanks)	30	Generating detailed technical drawings and maps (GIS/CAD) for all physical assets, including pumping stations and water storage structures.
2.5	Integration design between IT and civil systems	14	Creating interface documents and protocols to ensure the SCADA/IoT systems integrate seamlessly with physical civil assets (e.g., control valve actuators, flow meters).
2.6	Design review and approvals	7	Formal review of all design outputs by the PMC, regulatory bodies, and internal engineering teams before moving to procurement.

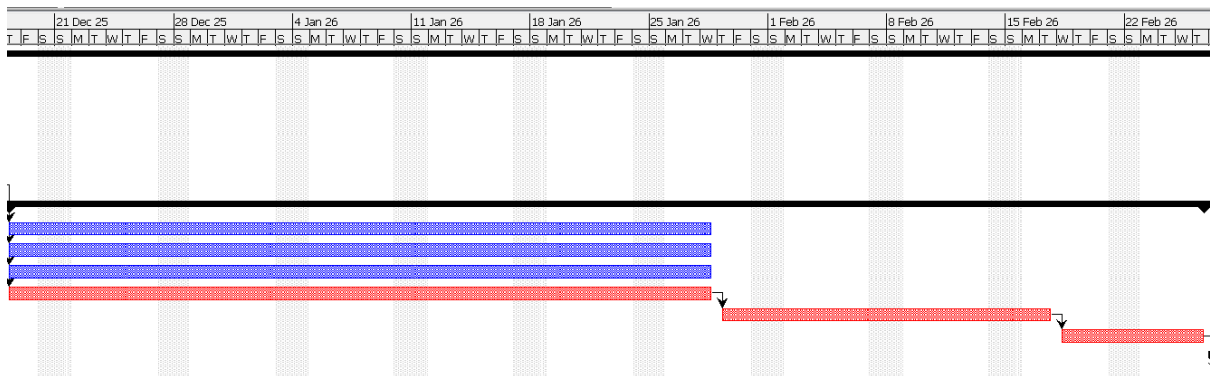


Figure 2: Phase-2.0 Breakdown

Phase 3.0: Procurement

- WBS ID: 3.0
- Duration: 52 Days
- Start Date: February 27, 2026
- Finish Date: May 11, 2026

- Objective: To finalize vendor contracts and procure all necessary physical and digital equipment based on approved designs. This phase is critical and links directly to construction readiness.

WBS ID	Task Name	Duration (Days)	Key Activities & Deliverables
3.1	Prepare Bill of Quantities (BoQ)	24	Finalizing the comprehensive list of materials and quantities required (pipes, meters, valves, hardware).
3.2	Vendor selection and tendering	56	Executing the tendering process and selecting suppliers for both the civil and IT components.
3.3	Procurement of pipes, meters, sensors, SCADA hardware	224	Issuing purchase orders and managing the logistics for delivery of major physical components.
3.4	Procurement of IT infrastructure (servers, network, security)	224	Acquiring necessary servers, communication equipment, and cybersecurity tools for the central control centre.
3.5	Quality inspection and delivery management	112	Inspecting incoming materials against specifications and coordinating site delivery to align with the construction schedule.

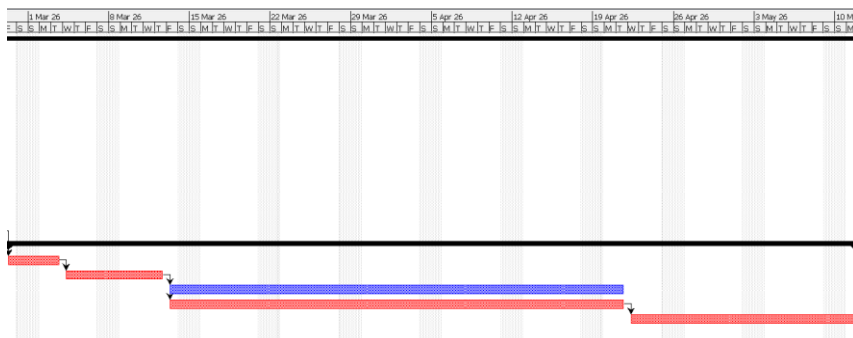


Figure 3: Phase-3.0 Breakdown

Phase 4.0: Construction & Installation

- WBS ID: 4.0
- Duration: 1200 Hours (150 Days)
- Start Date: May 12, 2026
- Finish Date: December 7, 2026
- Objective: The physical implementation phase, which poses the highest risk due to public and traffic constraints.

WBS ID	Task Name	Duration (Days)	Key Activities & Deliverables
4.1	Excavation and pipe laying	30	Digging trenches and laying the primary and secondary distribution pipelines across Zone-III.
4.2	Construction of reservoirs, valve chambers, and pump houses	90	Building new and upgrading existing concrete structures required for water storage, pressure regulation, and pumping operations.
4.3	Installation of distribution lines and connections	60	Installing local distribution lines and making the final connection points for household service lines.
4.4	Installation of smart water meters at households	30	The critical task of installing 150,000 smart meters, requiring significant coordination with residents and the community officer.
4.5	Road restoration and site cleanup	30	Restoring all affected roads, pavements, and public areas to their original or improved condition.

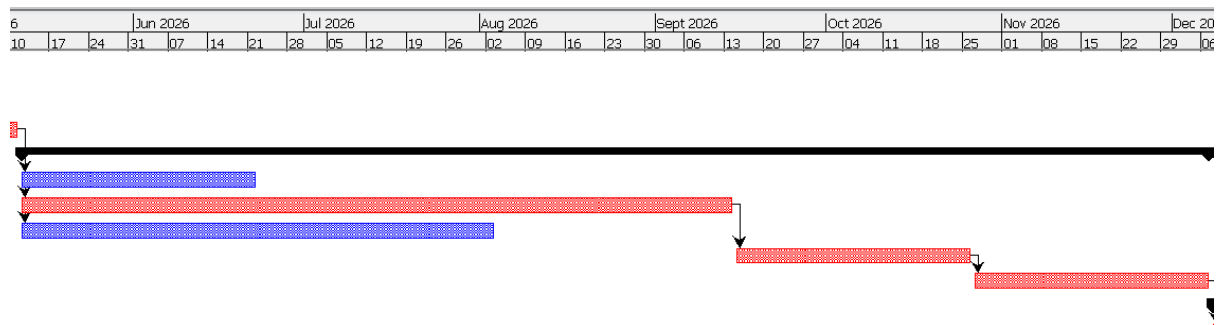


Figure 4: Phase-4.0 Breakdown

Phase 5.0: System Integration (IoT & SCADA)

- WBS ID: 5.0
- Duration: 720 Hours (90 Days)
- Start Date: December 8, 2026
- Finish Date: April 12, 2027
- Objective: To activate the "smart" functionality by integrating the physical field devices (Phase 4.0) with the digital control centre.

WBS ID	Task Name	Duration (Days)	Key Activities & Deliverables
5.1	Configuration of IoT sensors and smart meters	30	Programming 150,000 meters and field sensors to transmit data and correctly setting up communication modules.
5.2	SCADA software setup and dashboard creation	30	Installing, testing, and customizing the SCADA software platform and designing the real-time operational dashboard for utility staff.
5.3	Data connectivity setup (communication networks)	45	Establishing secure, robust communication networks (e.g., fibre, wireless backbone) between the control centre and all field devices.

WBS ID	Task Name	Duration (Days)	Key Activities & Deliverables
5.4	Integration of SCADA, IoT, and billing systems	30	Creating API links and protocols to ensure meter data automatically feeds into the new billing system.
5.5	Cybersecurity and data privacy setup	30	Implementing security protocols, intrusion detection, and establishing data privacy compliance for all collected customer and operational data.

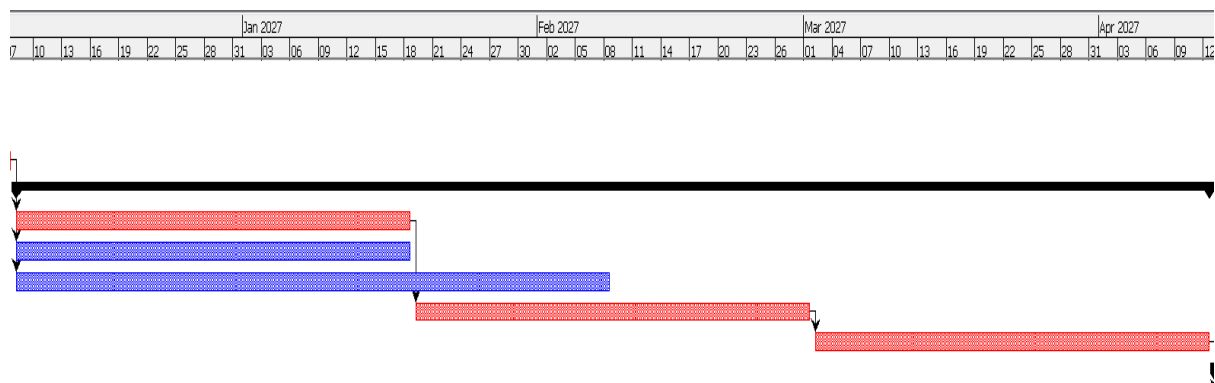


Figure 5: Phase-5.0 Breakdown

Phase 6.0: Testing & Commissioning

- WBS ID: 6.0
- Duration: 616 Hours (77 Days)
- Start Date: April 13, 2027
- Finish Date: July 28, 2027
- Objective: Rigorously verify that the physical infrastructure and the digital systems meet the performance and quality standards required for 24x7 service.

WBS ID	Task Name	Duration (Days)	Key Activities & Deliverables
6.1	Pressure and leakage testing of pipelines	14	Hydrostatic pressure testing of new pipelines and initial leak detection runs using acoustic sensors.
6.2	Calibration and testing of smart meters	14	Verifying the accuracy and reliability of the 150,000 installed smart meters.
6.3	SCADA and IoT system performance testing	14	Stress-testing the digital system under simulated high-load conditions to ensure stability and latency requirements are met.
6.4	Pilot run in select areas	28	Running the system with live water for a sustained period in selected zones to validate the 24x7 supply and NRW reduction goals.
6.5	Final approval and system handover	7	Formal sign-off on the commissioning report by the PMC and project sponsor, marking readiness for the transition phase.

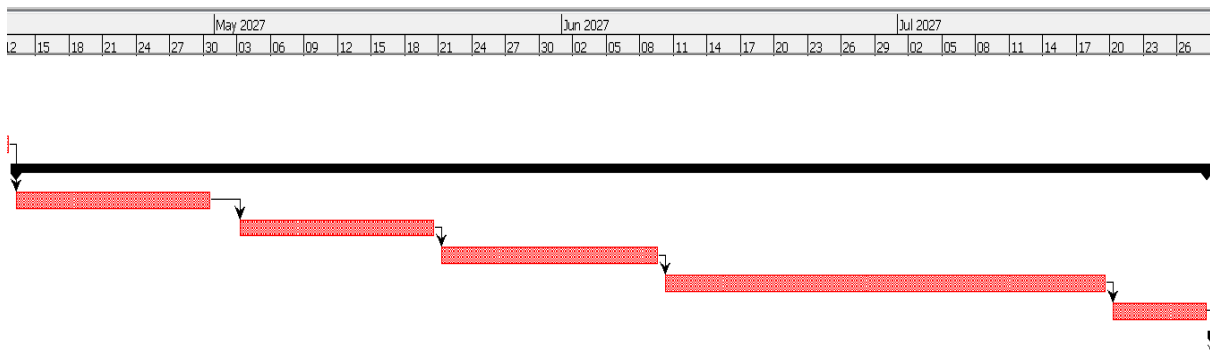


Figure 6: Phase-6.0 Breakdown

Phase 7.0: Training & Handover

- WBS ID: 7.0
- Duration: 1440 Hours (180 Days)
- Start Date: July 29, 2027
- Finish Date: April 5, 2028
- Objective: To ensure the PMC staff, operators, and the public are ready to use and maintain the new system.

WBS ID	Task Name	Duration (Days)	Key Activities & Deliverables
7.1	Training of PMC staff and operators on SCADA and IoT systems	90	Comprehensive training sessions for field technicians and control room operators on managing the new smart water network.
7.2	Documentation and operation manuals	60	Creating and validating all technical and operational manuals for future maintenance and troubleshooting.
7.3	Handover of system and assets to PMC	30	Formal transfer of ownership, documentation, and operational responsibility for all installed assets.
7.4	Citizen awareness programs and feedback collection	60	Public campaigns to educate the 150,000 households on using the new meters, reading bills, and understanding the 24x7 supply model.

Phase 8.0: Monitoring & Evaluation

- WBS ID: 8.0
- Duration: 1072 Hours (134 Days)
- Start Date: October 21, 2027
- Finish Date: April 25, 2028

- **Objective:** Post-implementation review, ensuring long-term operational stability and confirming the realization of project benefits. This phase overlaps with Phase 7.0.

WBS ID	Task Name	Duration (Days)	Key Activities & Deliverables
8.1	Continuous monitoring of water flow and leak detection	30	Initiating and maintaining real-time tracking of NRW indicators and automatic leak alerts.
8.2	Monthly performance reporting and KPI tracking	60	Regular generation of reports detailing water usage, NRW reduction metrics, and system uptime against the project goals.
8.3	Maintenance scheduling and system updates	30	Establishing the new preventative and predictive maintenance schedule based on IoT data and implementing initial system patches.
8.4	Audit of system efficiency and citizen satisfaction	30	Conducting independent audits to confirm the realized efficiency gains and gathering formal feedback from the 150,000 households.
8.5	Final evaluation report submission	14	Compiling the final project report, detailing goal achievement, lessons learned, and the final status of all deliverables.

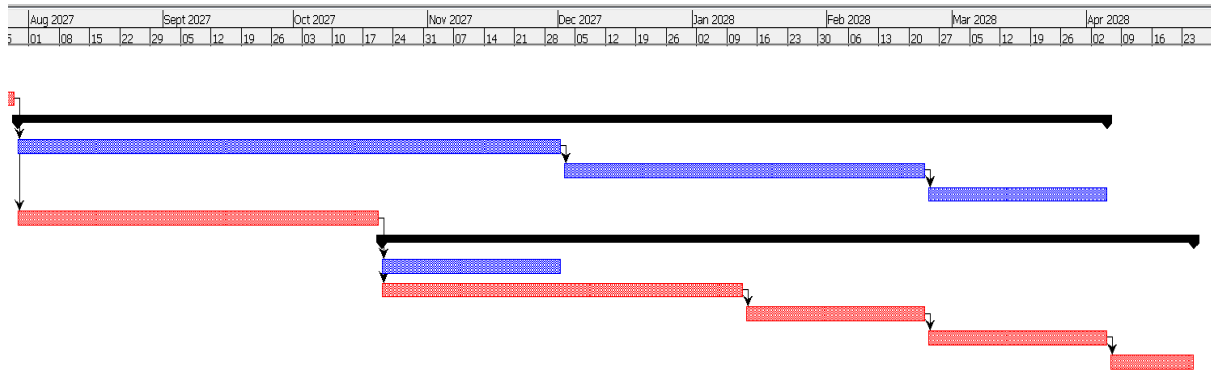


Figure 7: Phase-7 & 8 Breakdown

3. Project communication, Procurement & Implementation Plan:

The communication, procurement, and implementation planning and management strategies for Zone-III, Pune's Smart City Water Supply and Distribution Project are described in this report. This project's primary objective is to use SCADA and IoT technologies to give roughly 150,000 households a dependable water supply around-the-clock. This section describes how we will address data integration, coordination, and cybersecurity concerns throughout the project lifecycle, as well as how we will manage vendor and procurement processes and stakeholder communication.

3.1 Communication management plan:

A large-scale project like the Smart City Water Supply initiative, which involves numerous departments, contractors, vendors, and community members, requires an efficient communication plan. Accountability is ensured, delays are decreased, and misunderstandings are avoided with clear and consistent communication. Different kinds of information must be provided to each stakeholder at the right frequencies and through particular channels.

Stakeholder	Role/Interest	Information Required	Communication Method	Frequency	Responsible Person
Pune Municipal Corporation (PMC)	Project Sponsor & Regulator	Progress reports, financial utilization, risks	Formal progress reports, review meetings	Monthly	Project Director
Project Manager (S. Niteswara Sai)	Project Execution Lead	All project updates and integration reports	Daily internal updates, dashboards	Weekly	All team leads

Stakeholder	Role/Interest	Information Required	Communication Method	Frequency	Responsible Person
Civil Contractors	Construction execution	Design drawings, schedule updates	On-site coordination meetings, emails	Weekly	Civil Engineer
IoT & SCADA Vendors	System integration and hardware suppliers	Technical specs, installation schedule	Technical workshops, online calls	Bi-weekly	IoT Engineer
Cybersecurity Team	Data protection and system integrity	Network architecture, access logs	Secure portal updates	Monthly	Cybersecurity Lead
Citizens (150,000 households)	End users	Service updates, awareness on smart meters	Public meetings, media releases	Quarterly	Communication Officer
Central & State Govt. Departments	Funding and policy support	Status, compliance, fund utilization	Official reports	Quarterly	PMC Liaison Officer

3.2 Procurement Strategy and Vendor Management:

Procurement is also one of the most important elements of the project since it guarantees that all civil, electrical, and IoT elements are procured, delivered, and installed within schedule. The procurement strategy emphasizes quality, cost effectiveness, and vendor reliability. Procurement will be carried out following a centralized model under the administration of PMC and the Project Management Consultant.

The vendors will be chosen by an open tendering process based on technical competence, past experience in municipal or smart city projects, and meeting BIS and ISO quality standards. The major procurement categories are civil materials, smart meters, sensors, data servers, SCADA hardware, and cybersecurity solutions.

Vendor management will be ensured by explicit Service Level Agreements (SLAs), which specify timelines, payment milestones, and performance metrics. Vendor performance will be tracked regularly using Key Performance Indicators (KPIs) like punctuality of delivery, rate of defect, and response time to support. Any deviation from terms of contract will be evaluated by the procurement committee.

3.3 Role of IoT and SCADA Systems:

The convergence of SCADA and IoT constitutes the technical infrastructure of this initiative. Internet of Things (IoT) devices like smart water meters, pressure sensors, and leak detectors will be installed throughout the distribution network to gather real-time information.

The SCADA (Supervisory Control and Data Acquisition) system will act as the main control center. The SCADA system will collect data from IoT devices, track pressure and flow conditions, and enable operators to remotely control pumps, valves, and treatment systems. Real-time control enhances response time, minimizes water wastage, and helps maintain a consistent supply.

For instance, when a leakage sensor realizes there is unusual flow in a particular section, the data from IoT generates an alert at the SCADA interface. The operator can instantly shut down neighboring valves or send maintenance teams. This amount of automation maximizes operational efficiency and reduces non-revenue water loss.

3.4 Coordination Challenges Between IT and Civil Services:

Coordination between civil contractors who carry out physical construction and the IT teams involved in system integration is one of the biggest challenges of deploying a smart water system. Civil works like laying pipes and chamber construction need to be coordinated with the deployment of IoT sensors and network cabling. Civil work delays tend to affect IT deployment timelines.

In addition, technical mismatches may happen when IoT devices are not in sync with the physical layout or design. To avoid this, civil and IT engineers will meet for joint planning every week. Digital tools such as Building Information Modeling (BIM) and GIS mapping will be used to ensure shared and accurate data from the project for both teams.

There will be a sole Integration Committee, headed by the Project Manager, with responsibility for tracking cross-domain dependencies and approving the IT team's takeover of completed civil sections. This formal coordination reduces rework and ensures that project timelines are kept intact.

3.5 Data Integration and Cybersecurity Risk mitigation:

Data integration is essential in the Smart Water Supply Project since it will ensure that data gathered from IoT sensors, SCADA systems, and billing platforms are harmonized and available for decision-making. A centralized data platform will be used to manage data exchange between the various systems via secure APIs. This will support precise monitoring of water flow, consumption patterns, and revenue tracking.

Cybersecurity is equally critical with the reliance on interconnected digital infrastructure. Intruder access or malware attacks can cause service interruptions or undermine sensitive information. To address these risks, the project will implement multi-layered security measures, such as encrypted data transmission, role-based access control, and periodic system audits.

Routine threats in the form of unauthorized access to SCADA, IoT breaches, and ransomware will be mitigated via preventive strategies in the form of network segmentation, firewall mechanisms, and regular penetration testing. A cybersecurity professional will ensure adherence to ISO/IEC 27001 standards and CERT-In regulations for continuous monitoring and system resilience.

The Smart City Water Supply Project of Zone-III, Pune, is intended to deliver a 24x7 assured water supply to 150,000 homes through advanced IoT and SCADA technologies. The project emphasizes effective communication, speedy procurement, and hassle-free coordination among all teams.

A good communication plan ensures that all stakeholders receive the correct information at the right time, enhancing teamwork and responsibility. The procurement process is created to choose quality vendors based on open bidding and follow up with them on performance against clear goals.

IoT and SCADA systems facilitate real-time monitoring, prompt leak detection, and more effective water flow control. Timely coordination between IT and civil teams avoids delays and smooth installation. Effective cybersecurity practices and secure data systems safeguard the project against cyber threats.

4.1 Project Monitoring and Controlling Plan

Control of the project concentrated on maintaining schedule and cost integrity in all phases of the Work Breakdown Structure (WBS), particularly high-risk civil and procurement works.

Progress Measurement and Analysis

Daily Earned Value Management (EVM) principles were consistently applied, with weekly measurement of Schedule Variance and Cost Variance. Project Libre schedule data confirmed that resource levelling and dependency management were achieved for the Planning and Design phase (WBS 1.1 and 1.2). Significant variations did occur in the Construction and Installation phase (WBS 1.4), primarily due to external constraints described below.

Corrective and Preventive Actions (CAPA)

In order to counteract slippage, Corrective and Preventive Actions (CAPA) were ordered immediately, focusing on acceleration of the critical path:

1. Turbo-Charging Integration: The integration stage (WBS 1.5) was turbo-charged by overlapping technology deployment (SCADA/IoT installation) with the subsequent stages of civil work. This maximized deployment of the IT and SCADA engineering team.
2. Acceleration of Resources: The Welders and Pipe Fitters, the primary human resources, were authorized to go on overtime and double shifts in order to reduce the period of pipeline installation (WBS 1.4.1), recovering approximately 60% of lost time.
3. Forefront Risk Management: Hand-in-hand with the Communication & Community Officer (CCO), Site Supervisors identified and cleared local grievances before they turned into formal protests and hence prevented additional schedule slippage.

4.2 Quality Control and Testing Procedure

Quality control was of top priority, so that all technology and civil elements met high operating standards as set out in the technical specifications.

Civil Infrastructure Quality Checks

- Pipeline Integrity: The most basic Quality Control task for pipelines (WBS 1.4.1) was Pressure and Leakage Testing (WBS 1.6.1). \$1.5\$ times the maximum operating pressure for 24 hours was applied to all the sections and measured {Leakage Ratio}, hence confirming successful installation and joint integrity.
- Structural Verification: Thorough Structural Integrity Inspections were also performed on the primary infrastructure like pump houses and reservoirs (WBS 1.4.2). Concrete strength tests (28-day cube tests) ascertained compliance with the prescribed M30 requirement, linking physical construction with approved civil drawings (WBS 1.2.4).

Smart Meter and SCADA Commissioning

- **Smart Meter Calibration:** The Meters (WBS 1.4.4 & 1.6.2) underwent rigorous Calibration and Accuracy Tests. Deviation from accuracy was guaranteed to be within permissible tolerance against primary reference standards.
- **System Performance:** The SCADA and IoT integrated systems (WBS 1.6.3) achieved the operational objectives, displaying ultra-low command latency and system redundancy. The system reached a steady rate of data packet delivery at 99%.
- **Cybersecurity Compliance:** Cybersecurity Experts performed a final check on the data communication and control systems (WBS 1.5.5), assuring that all high-severity security bugs were resolved prior to final handover.

4.3 Delays Due to Public Protests and Road Closures – Mitigation

Unforeseen public opposition was the most significant schedule risk in implementation.

Impact Assessment

Surprise community opposition and delayed municipal approvals of street closings resulted in a critical path slip of approximately 45 days in the Excavation and Pipe Laying phase (WBS 1.4.1). The slip stalled deployment of heavy equipment and civil teams.

Mitigation Strategies Implemented

The response was directed at a combination of operational adjustments and community relations:

- **Community Relations (WBS 1.7.4):** The CCO introduced increased local publicity campaigns, shifting public opinion and facilitating increased access to partially restricted working areas, which was key to securing vital work fronts.
- **Segmented Work Approach:** Large pipeline contracts were segmented into small, nimble "Segmented Sprints." This channeled legal and public access risks, so political issues within a block would not halt work across the board all along Zone-III.
- **Authorization for Night Work:** Altered Traffic Management Plans facilitated nighttime work in high-density commercial areas, minimizing day-time traffic disturbance and accelerating work during non-working hours.

4.4 Lessons Learned and Recommendations

Project performance analysis determines lessons for improvement for following phases (Zone-IV, V, etc.).

Learnings

- **Risk Priority of Stakeholders:** CCO activity of community and political engagement must be elevated to a Tier 1 Risk priority, including loading resources by the CCO at an earlier stage in the WBS (WBS 1.1).

- Procurement Buffer: Application of worldwide supply chains for extremely specialized IoT equipment nearly caused considerable delay. Advance commitment towards developing a physical material reserve (WBS 1.3.3) of major items is necessary to counteract market trends.
- Design Synchronization: Synchronized coordination among Civil and IT Design groups (WBS 1.2.4/1.2.2) prevented serious design clashes at execution level. Such collaborative design model must be applied rigorously.

Recommendations for Future Phases (Zone-IV, V, etc.)

1. Proactive Community WBS: Include a mandatory 60-day Community Relations Phase prior to starting WBS 1.4.
2. Contractual Modularity: Implement brief, milestone-based construction contracts to improve contractor accountability and operational responsiveness.
3. Digital Integration Improvement: Future releases of the system software should have integrated WBS 1.8.1 Non-Revenue Water computation logic for real-time leak detection and move beyond reactive data analysis to predictive maintenance.

4.5 Conclusion

The SMART CITY WATER SUPPLY Modernization project in Zone-III will be an outstanding engineering achievement. By virtue of successful technical execution and adaptive response to unforeseen social challenges, Elite Hydro Constructions delivers a modern, high-level, and functionally efficient water infrastructure. The experience gained in risk mitigation and stakeholder management provides a good foundation for the assurance of the successful and efficient rollout of future Smart City zones.

References:

- ☐ Government of India. (2015). *Smart Cities Mission: Statement and Guidelines*. Ministry of Housing and Urban Affairs. <https://smartcities.gov.in>
- ☐ Pune Municipal Corporation. (2023). *Smart Water Supply Implementation Report*. <https://pmc.gov.in>
- ☐ Project Management Institute. (2021). *A Guide to the Project Management Body of Knowledge (PMBOK® Guide) (7th ed.)*. Project Management Institute, Inc.
- ☐ Kumar, S., & Gupta, R. (2021). *Role of IoT in Smart Water Supply Management Systems*. *International Journal of Smart City Applications*, 3(1), 27–39.
- ☐ Aithal, A., & Aithal, P. S. (2021). *Urban Water Governance in India: Challenges and Smart Solutions*. *International Journal of Case Studies in Business, IT, and Education*, 5(2), 122–134.
- ☐ Chourasia, V., & Dutta, S. (2020). *Optimization of Water Distribution Networks Using GIS and SCADA Integration*. *Water Resources Management*, 34(12), 3987–4003.
- ☐ Deloitte India. (2021). *Smart Cities: Transforming Urban Governance through Technology*. Deloitte Insights.